

CLINICAL FINDINGS

History

The optimal evaluation of a patient with suspected graft-versus-host disease (GVHD) begins with a thorough review of key historical factors. Important elements include the type of stem cell transplantation (allogeneic or autologous), any manipulation of stem cells prior to infusion, the conditioning (preparative) regimen, the date of stem cell infusion, white blood cell count recovery, medications administered (including immunosuppressants and antimicrobials), and the overall hospitalization course.

Although acute cutaneous GVHD most commonly occurs after allogeneic stem cell transplantation, it can also develop following maternal-fetal lymphocyte transfer, blood transfusion, and solid organ transplantation—particularly liver transplantation, due to the organ's high content of immunocompetent lymphocytes.

Extracutaneous manifestations such as diarrhea, hepatitis, or gastrointestinal symptoms may support the diagnosis. However, early GVHD often presents with isolated skin involvement, so the absence of systemic symptoms should not exclude the diagnosis. Correlation of clinical history with cutaneous and histopathologic findings is essential for accurate diagnosis of acute or chronic GVHD, enabling timely intervention.

Cutaneous Lesions

Acute Cutaneous Graft-Versus-Host Reaction After Allogeneic Transplantation

The cutaneous eruption of acute graft-versus-host reaction (GVHR) typically begins with faint erythematous macules that appear on any skin surface, but

most commonly affect the **pinnae, palms, and soles** (stage 1). These early lesions are often symmetric and may be subtle.

The presence of palmoplantar erythema is frequently noted, though its specificity is limited, as chemotherapy alone (without stem cell support) can cause similar acral erythema. Edema and tenderness commonly accompany the erythematous areas, regardless of etiology. Pruritus is generally mild or absent.

As the condition progresses, the erythematous macules increase in number and coalesce into larger confluent patches, frequently involving the **upper back and trunk** (stages 2–3). In early stages, lesions on the trunk may exhibit a **perifollicular distribution**, which can be a helpful diagnostic clue.

With further evolution, widespread erythema may lead to **erythroderma** (involvement of >90% of body surface area), which is often faint and diffuse. The development of **bullae** (stage 4) signifies a poor prognosis. These bullous lesions are **subepidermal** with **necrotic epidermal roofs** and closely resemble **toxic epidermal necrolysis (TEN)** clinically and histologically.

A staging system for acute cutaneous GVHR is commonly used to assess severity and guide management.

The progression from initial macules to erythroderma or bullae typically unfolds over several weeks. A rare **explosive variant** has been described, characterized by rapid onset of erythroderma followed by swift transition to a lichenoid eruption resembling chronic GVHD. The risk factors for this aggressive form remain unclear.

With effective treatment, the skin eruption resolves in a reverse sequence to its appearance, often accompanied by **desquamation**, particularly in areas of prior erythema or blistering.